Rotational Lorentz Contraction as the Geometric Origin of the Anomalous Magnetic Moment: A Structural Correspondence with the Muon Lifetime Problem

(Structural Clarification Note)

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We demonstrate that the anomalous magnetic moment of the electron a_e and the muon atmospheric lifetime extension are two manifestations of the same relativistic length-extension mechanism. In the 0-Sphere electron model, Thomas precession applied to sinusoidal internal oscillation yields the double-angle angular velocity $\Omega \propto \sin(2\omega t)$, geometrically necessitating $g = 2$. The deviation from $g = 2$ is identified with the rotational Lorentz contraction of the internal orbital arc $2\pi r \in [L]$: the arc is shortened by the factor $1/(1 + a_e)$ in the laboratory frame, in structural analogy to the extension of the muon's proper path length $L_0 \in [L]$ to $\gamma_L L_0$. An inverse calculation from $a_e^{\rm exp}$ yields the Zitterbewegung velocity $v_{\rm ZB} \approx 0.04c$ and the Lorentz factor $\gamma_L = 1 + a_e$. In both cases, a length $[L]$ is modified by the Lorentz factor of the relevant velocity; the structural mechanism is identical. The forward derivation—determining $v_{\rm ZB}$ independently of a_e from Thomas precession alone—remains the open task whose completion would elevate this correspondence to a confirmed physical equivalence.

I. INTRODUCTION

The anomalous magnetic moment of the electron, $a_e = (g-2)/2 \approx 1.16 \times 10^{-3}$ [1], is among the most precisely measured quantities in physics. Its standard theoretical treatment invokes multi-loop quantum electrodynamics, a procedure that is highly successful numerically but provides little geometric intuition for why such a deviation from the Dirac prediction $g = 2$ exists at all.

A separate, well-established phenomenon of special relativity is the atmospheric muon surplus: muons created by cosmic rays at altitudes of ∼15 km reach the ground at rates far exceeding what their proper lifetime $\tau_0 \approx 2.2\,\mu$s would allow [2]. The resolution is textbook: time dilation extends the laboratory lifetime to $\tau_{\rm lab} = \gamma_L \tau_0$, equivalently the laboratory path length to $L_{\rm lab} = \gamma_L L_0$, where $\gamma_L = 1/\sqrt{1 - v^2/c^2} \gg 1$ for $v \approx 0.9994c$.

This paper identifies a structural correspondence between these two phenomena. Both involve a length $[L]$ modified by a Lorentz factor: the muon path length L_0 extended by $\gamma_L \gg 1$, and the electron's internal orbital arc $2\pi r$ contracted by $1/\gamma_L$ where $\gamma_L - 1 = a_e \approx 10^{-3}$. The mechanism is identical; only the velocity, and hence the magnitude of $\gamma_L - 1$, differs. This correspondence was identified through an inverse calculation in Ref. [5]; the present paper makes the structural analogy explicit and self-contained.

The remainder of this paper is organized as follows. Section II establishes the general relativistic length-extension identity $\Delta[L] = |\gamma_L - 1| \cdot L_0$ that underlies both phenomena. Section III reviews the muon lifetime experiment as the textbook realization of this identity at $v \approx c$. Section IV develops the 0-Sphere Model account of the electron: Sec. IV A derives the double-angle angular velocity from Thomas precession with sinusoidal oscillation, Sec. IV B identifies rotational Lorentz contraction of the orbital arc $2\pi r \in [L]$ as the geometric source of a_e, and Sec. IV C carries out the inverse calculation yielding $v_{\rm ZB} \approx 0.04c$ and $\gamma_L = 1 + a_e$. Section V places the two phenomena side by side in a unified table and figure. Section VI states the open forward chain whose completion would elevate the present correspondence to a confirmed equivalence. Section VII summarizes the main result.

II. RELATIVISTIC LENGTH EXTENSION: GENERAL STRUCTURE

Special relativity assigns to any object moving at velocity v a Lorentz factor

$$\gamma_L = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{II.1}$$

A length $[L]$ associated with the rest frame of that object is observed in the laboratory frame to be modified by γ_L. For an object in motion, its proper path length L_0 is extended to

$$L_{\rm lab} = \gamma_L \cdot L_0, \tag{II.2}$$

while a transverse length (such as a rotating circumference in the direction of motion) is contracted to

$$L_{\rm lab} = \frac{L_0}{\gamma_L}. \tag{II.3}$$

In both cases the surplus or deficit is

$$\Delta[L] = |\gamma_L - 1| \cdot L_0, \tag{II.4}$$

a quantity that vanishes for $v \to 0$ and grows with v. The velocity v sets the scale; the algebraic structure is universal.

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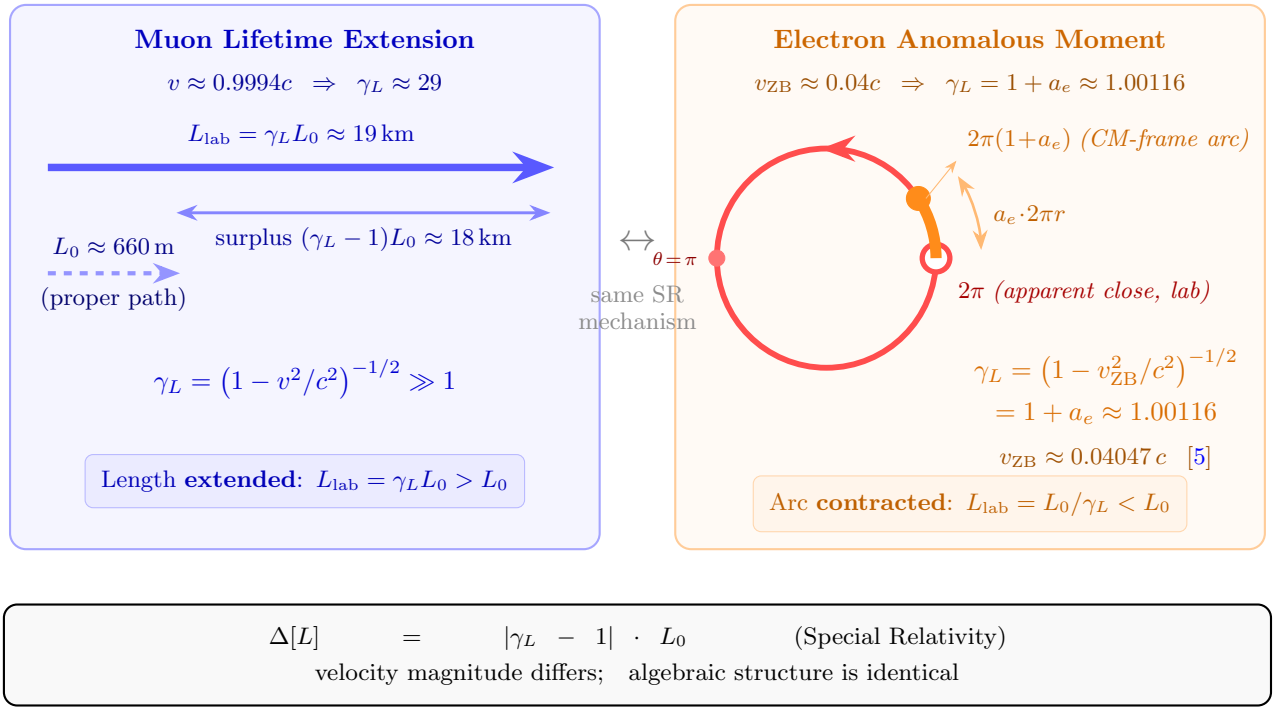

Fig. 1. Structural correspondence between the muon lifetime experiment and the electron anomalous magnetic moment in the 0-Sphere Model

Left (blue): A muon traveling at $v \approx c$ has $\gamma_L \gg 1$; its laboratory path length $L_{\rm lab} = \gamma_L L_0$ far exceeds the proper path L_0, allowing it to reach the ground. **Right (orange):** The electron's internal orbital arc $2\pi r$ is shortened by $1/(1+a_e)$ in the laboratory frame. The orange arc indicates the excess $a_e \cdot 2\pi r$ (exaggerated for visibility). In both cases, a length $[L]$ is modified by the Lorentz factor γ_L of the relevant velocity.

III. MUON LIFETIME: THE TEXTBOOK CASE

Muons produced by cosmic-ray collisions at altitude $h \approx 15$ km travel at $v \approx 0.9994c$, giving $\gamma_L \approx 29$. Their proper lifetime $\tau_0 \approx 2.2\,\mu$s would imply a maximum path length of $L_0 = v\tau_0 \approx 660$ m, far short of 15 km. The experimental observation that a large fraction reaches the ground [2] is explained by Eq. (II.2):

$$L_{\rm lab} = \gamma_L \cdot L_0 \approx 29 \times 660 \text{ m} \approx 19 \text{ km}. \tag{III.1}$$

The surplus path $(\gamma_L - 1)L_0 \approx 18$ km is the directly observable consequence of $\gamma_L \gg 1$.

This is the paradigmatic SR length-extension event: a proper length $L_0 \in [L]$, extended by γ_L of the relevant velocity. No internal structure of the muon is required; the mechanism is purely kinematic.

IV. ELECTRON INTERNAL ORBIT: THE 0-SPHERE MODEL

A. Thomas Precession with Sinusoidal Oscillation

The 0-Sphere electron model [3] proposes that two discrete energy kernels A and B exchange thermal potential energy via an internal photon oscillating sinusoidally. Let the photon velocity and acceleration be

$$\boldsymbol{v}_{\gamma^*} = \cos\omega t, \tag{IV.1}$$
$$\boldsymbol{a}_{\gamma^*} = -\sin\omega t. \tag{IV.2}$$

Thomas precession [4] gives the angular velocity of a coordinate frame co-moving with an accelerated particle as observed from the laboratory:

$$\boldsymbol{\Omega} = \frac{1}{2c^2}\,[\boldsymbol{a} \times \boldsymbol{v}], \tag{IV.3}$$

where the approximation $\beta = 1 - v^2/c^2 \fallingdotseq 1$ (non-relativistic internal velocity) is applied. Substituting Eqs. (IV.1)–(IV.2) into Eq. (IV.3):

$$\begin{aligned}\boldsymbol{\Omega} &= \frac{1}{2c^2}\,[-\sin\omega t \times \cos\omega t] \\ &= -\frac{1}{4c^2}\,\sin 2\omega t.\end{aligned} \tag{IV.4}$$

The double-angle structure $\sin 2\omega t$ is the key result: the angular velocity oscillates at twice the frequency of the displacement. One period of internal oscillation (ωt) produces two cycles of angular velocity ($2\omega t$), which is the geometric basis for $g = 2$—the electron's spin generates a magnetic field twice as efficiently as an orbital current [5–7].

Table I. Structural comparison of muon lifetime extension and electron anomalous magnetic moment

Property	**Muon lifetime (CERN)**	**Electron a_e (0-Sphere)**
Length modified $[L]$	Proper path length $L_0 = v\tau_0$	Internal orbital arc $L_0 = 2\pi r$
Relevant velocity	$v \approx 0.9994\,c$	$v_{\mathrm{ZB}} \approx 0.04047\,c$
Lorentz factor γ_L	≈ 29 (extension $\gg 1$)	$\approx 1 + 1.16 \times 10^{-3}$ (contraction $\ll 1$)
Observable surplus / deficit	$(\gamma_L - 1)L_0 \approx 18$ km (path surplus)	$a_e \cdot 2\pi r$ (arc shortening)
Observed quantity	Muon flux at ground level [2]	Anomalous magnetic moment a_e [1]
Calculation direction	Forward: v known $\to \gamma_L \to L_{\mathrm{lab}}$	Inverse (current): a_e known $\to v_{\mathrm{ZB}}$ [[5]]
Open forward direction	(completed)	Thomas precession $\to v_{\mathrm{ZB}} \to a_e$ (future)

Both phenomena are governed by Eq. (II.4): $\Delta[L] = |\gamma_L - 1| \cdot L_0$. The velocity magnitude sets the scale of the effect; the algebraic structure is identical. The RMS correction in Eq. (IV.6)—introduced by the present author in Ref. [5] and not a standard textbook result—arises because the harmonic oscillator produces peak rather than average Lorentz contraction; the muon case requires no such correction because v is constant.

Furthermore, since $\sin 2\omega t$ takes both positive and negative values, the angular velocity reverses sign within a single oscillation period. Whether this sign alternation maps directly onto the spin-up/spin-down dichotomy—and whether such a mapping can be made without invoking quantum superposition—remains under active investigation within the 0-Sphere framework. The factor $\frac{1}{2}$ appearing in the Thomas angular velocity $\Omega_T \approx \frac{1}{2}(\mathbf{v}/c^2) \times \mathbf{a}$ is one of three independent occurrences of $\frac{1}{2}$ in the 0-Sphere Model—the others being a geometric centroid constraint and the spin quantum number $s = \frac{1}{2}$ as a topological invariant—whose common origin will be examined in a future paper. Notably, the factor 2 in the denominator of the definition $a_e = (g-2)/2$, the factor 2 in the double-angle $\sin 2\omega t$ of Eq. (IV.4), and the factor $\frac{1}{2}$ in Ω_T are three expressions of the same underlying structure: a single internal oscillation cycle generates two cycles of angular velocity, so that π of spatial arc suffices to produce 2π of magnetic response.

B. Rotational Lorentz Contraction and the Anomalous Moment

In the non-relativistic limit, the factor ($\beta \fallingdotseq 1$) in Eq. (IV.3) sets the rotational Lorentz contraction to zero, recovering the ideal $g = 2$ result. However, if the internal oscillation velocity is non-negligible compared to c, the circumference in the direction of rotation is shortened by the Lorentz factor:

$$\frac{L}{L_0} = \frac{1}{\gamma_L} \approx \frac{1}{1 + a_e}, \tag{IV.5}$$

where $L_0 = 2\pi r$ is the uncontracted orbital arc (one full traversal $= \pi$ of internal phase, since Ω has period π) and a_e quantifies the fractional shortening. The rotational Lorentz contraction underlying this equation—the radius, being perpendicular to the velocity, undergoes no contraction, while the circumferential arc lies along the direction of motion and is shortened by γ_L—was identified by Einstein [11] and is the basis of the Ehrenfest paradox [10]. The present work applies this established mechanism to the electron's internal orbital arc as the geometric origin of a_e.

C. Inverse Calculation: $a_e \to v_{\mathrm{ZB}}$

Combining Eq. (IV.5) with the standard Lorentz contraction formula $L/L_0 = \sqrt{1 - v^2/c^2}$ gives [5, 8]

$$\sqrt{1 - \frac{v_{\mathrm{ZB}}^2}{c^2}} = \frac{1}{1 + \frac{1}{\sqrt{2}}\, a_e^{\mathrm{exp}}}, \tag{IV.6}$$

where the RMS factor $1/\sqrt{2}$ is introduced because the harmonic oscillator produces continuously varying acceleration, so a_e^{exp} corresponds to the maximum rather than the average Lorentz contraction (analogous to the RMS correction between peak and effective AC voltage). Substituting $a_e^{\mathrm{exp}} = 0.00115965\ldots$ yields

$$\frac{v_{\mathrm{ZB}}^2}{c^2} = 0.001638, \tag{IV.7}$$

$$v_{\mathrm{ZB}} \approx 0.04047\,c. \tag{IV.8}$$

This is the Zitterbewegung velocity of the internal photon within the 0-Sphere model. The Lorentz factor at this velocity is

$$\gamma_L = \frac{1}{\sqrt{1 - v_{\mathrm{ZB}}^2/c^2}} \approx 1 + a_e, \tag{IV.9}$$

confirming $\gamma_L = 1 + a_e$ as a structural consequence of SR applied to the internal orbital arc [8, 9].

Status of this result. The calculation above proceeds *from* a_e^{exp} *to* v_{ZB}: it is an inverse route that identifies the mechanism but does not constitute a forward derivation of a_e from first principles. The value $v_{\text{ZB}} \approx 0.04c$ is therefore presented as a *prediction* inferred from the measured anomalous moment, not as an independently observed quantity. This situation reflects a broader conventional position: Zitterbewegung is widely regarded as a mathematical artifact of the Dirac equation rather than a physical oscillation, and the canonical estimate places it at the speed of light c, rendering it unobservable. The 0-Sphere model departs from this view by proposing that ZB is a real, subluminal oscillation at $v_{\text{ZB}} \approx 0.04c$. Should a future experiment confirm this velocity directly—independently of the g-factor—the present one-directional inference would be superseded: Eq. (IV.6) would then function as a genuine two-way equation, allowing a_e to be derived from v_{ZB} and v_{ZB} to be derived from a_e on equal footing. The forward direction—deriving v_{ZB} from Thomas precession alone, independently of the measured value of a_e, and then obtaining a_e as output via Eq. (IV.9)—is the central open task of the 0-Sphere research programme.

V. STRUCTURAL COMPARISON

Figure 1 and Table I display the two phenomena side by side.

The shared structure can be stated precisely. In both cases:

(i) A length $L_0 \in [L]$ is defined in the rest or CM frame of the relevant object.

(ii) Special relativity modifies this length in the laboratory frame by the Lorentz factor γ_L of the relevant velocity v.

(iii) The surplus or deficit $\Delta[L] = |\gamma_L - 1| \cdot L_0$ is the directly observable quantity.

For the muon, $v \approx c$ and $\gamma_L - 1 \approx 28$; the effect is dramatic. For the electron, $v_{\text{ZB}} \approx 0.04c$ and $\gamma_L - 1 = a_e \approx 10^{-3}$; the effect is tiny but structurally identical.

VI. OPEN DIRECTION: THE FORWARD CHAIN

The result $v_{\text{ZB}} \approx 0.04c$ was obtained in Ref. [5] by inverse calculation from a_e^{exp}. This establishes that the mechanism is rotational Lorentz contraction of the arc $2\pi r \in [L]$, and identifies $\gamma_L = 1 + a_e$ as a structural consequence, but does not yet constitute a derivation of a_e from first principles.

The forward chain that would close the argument is:

$$\begin{aligned} \underbrace{\text{Thomas precession}}_{\text{Eq. (IV.4)}} \longrightarrow\ & \underbrace{v_{\text{ZB}}}_{\text{open task}} \\ & \text{(derived independently of } a_e\text{)} \\ \longrightarrow\ & \underbrace{\gamma_L = \left(1 - v_{\text{ZB}}^2/c^2\right)^{-1/2}}_{\text{SR, Eq. (II.1)}} \\ \longrightarrow\ & \underbrace{\gamma_L = 1 + a_e}_{\text{output: confirmed equivalence}} \quad . \end{aligned} \tag{VI.1}$$

At present, v_{ZB} has not been derived from the internal dynamics of the 0-Sphere model without reference to a_e^{exp}, nor has it been measured by an experiment independent of the g-factor. Either achievement—a first-principles calculation of the internal oscillation velocity, or a direct experimental measurement of v_{ZB}—would elevate the present structural correspondence to a confirmed physical equivalence between the anomalous magnetic moment and the muon lifetime experiment.

The Berry-phase and fiber-bundle treatment of the dual-frame structure, within which $\gamma_L = 1 + a_e$ appears as the bridge equation connecting the CM-frame and laboratory-frame g-factors, will be addressed in future work.

VII. CONCLUSION

We have shown that the anomalous magnetic moment of the electron and the muon atmospheric lifetime extension are two instances of the same relativistic length-extension identity $\Delta[L] = |\gamma_L - 1| \cdot L_0$. In the 0-Sphere model, Thomas precession with sinusoidal internal oscillation produces the double-angle angular velocity $\Omega \propto \sin 2\omega t$, establishing $g = 2$ as the geometric baseline. The deviation a_e is identified with the rotational Lorentz contraction of the internal orbital arc $2\pi r \in [L]$, and an inverse calculation from a_e^{exp} yields $v_{\text{ZB}} \approx 0.04c$, confirming $\gamma_L = 1 + a_e$.

The muon reaches the ground because $v \approx c$ makes $\gamma_L \gg 1$. The electron exhibits $a_e \approx 10^{-3}$ because $v_{\text{ZB}} \approx 0.04c$ makes $\gamma_L - 1$ small but nonzero. The physical mechanism is the same. The anomalous magnetic moment is, in this sense, the electron's version of the muon lifetime problem—a question about a length $[L]$ extended or contracted by the Lorentz factor of an internal velocity.

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The physical concepts, theoretical framework, and all original ideas presented in this work—including the

identification of the anomalous magnetic moment with rotational Lorentz contraction of the internal orbital arc, a central claim of the 0-Sphere model developed continuously since 2018—are solely the author's own. A large language model (LLM) was used exclusively for linguistic polishing and paragraph composition; it contributed no conceptual content.

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